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### Single pair connector module

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#### Abstract

A connector module is disclosed having physical dimensions for being installed into standardized, or pre-existing, structures such as a patch panel including pre-populated slot openings. The connector module provides simple field termination for different sized single pair cables using screw-down/cage-clamp for the bare single pair of conductors emerging from a centralized power equipment to be a pluggable mating connector terminating the field wires.

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**Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims benefit to U.S. Provisional Patent Application No. 63/272,774, filed Oct. 28, 2021, the entirety of which is

hereby incorporated by reference herein.

## FIELD OF THE TECHNOLOGY

(1) This disclosure relates to a connector module having a connector for receiving a single pair of wires (i.e., conductors) and an adapter housing for accepting the connector and having characteristic dimensions for fitting into existing patch panel ports. This connector module provides a simple and effective solution for field termination of a range of single pair cables.

## BACKGROUND

(2) Throughout the course of installing equipment within a structure, many different types of cables having different sized wire conductors may be used for different applications having different power rating requirements. For example, in the distributed antenna system (DAS) industry cable of various sizes may be used during an installation to distribute power from a centralized telecom or electrical closet in a building to antennas distributed throughout the interior of the building, which may consist of many stories. Conventional zone cabling enclosures do not provide a simple termination point for conductors having such a wide range of size and power ratings.

## SUMMARY

(3) This disclosure provides a simple field termination solution for single pair cable (i.e., cable including a single pair of wire conductors) within a range of sizes (e.g., 12-26 AWG) to utilize more standardized connector components.

(4) According to an embodiment, a connector module for installing into an equipment opening is provided. The connector module comprising an adapter configured to receive a connector jack, the adapter comprising: a top surface, a first side wall, a second side wall opposite the first side wall, a front side surface including an opening for receiving the connector jack, and a bottom surface, wherein at least a portion of the top surface, a portion of the first side wall, a portion of the second side wall, a portion of the front side surface, and a portion of the bottom surface form an internal cavity for housing the connector jack.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a connector module in an assembled state, according to an embodiment.
- (2) FIG. 2 is a perspective view of the connector module shown in FIG. 1 in a disassembled state to show an adapter and a connector that comprise the connector module.
- (3) FIG. 3 is a perspective view showing a bottom surface of the adapter and the connector illustrated in FIG. 2.
- (4) FIG. 4 is a rear perspective view showing an internal cavity of the adapter.
- (5) FIG. 5 is a front side view of the adapter.
- (6) FIG. 6 is a rear side view of the adapter.
- (7) FIG. 7 is a perspective view of the connector module preparing to receive a plug, according to some embodiments.
- (8) FIG. 8 is a perspective view of the plug inserted into the connector module, according to some embodiments.
- (9) FIG. 9 is a perspective view of a single pair cable installed into a plug, where the plug is inserted into the connector module that has been installed into a patch panel, according to some embodiments.

### DETAILED DESCRIPTION

(10) This disclosure describes embodiments of a connector module having physical dimensions for being installed into standardized, or pre-existing, structures such as a patch panel including pre-populated slot openings having characteristics for fitting the connector module. The connector

module disclosed herein is configured to provide simple field termination for different sized single pair cables (e.g., 12-26 AWG) using screw-down/cage-clamp for the bare single pair of conductors emerging from a centralized power equipment to be a pluggable mating connector terminating the field wires. A number of compatible, common-pitch-mating connectors can be snapped into a rear of rack-mountable panels once the field wiring conductors have been screwed down into the cage clamps. This gives a central and flexible (e.g., pluggable) head-end to field-wiring demarcation point that can greatly assist in overall installation and troubleshooting.

(11) FIG. 1 shows a connector module **100** comprising a connector **20** (e.g., connector jack) and an adapter **10**. In FIG. 1, the connector module **100** is shown to be in an assembled state where the connector **20** is fully inserted and secured to the adapter **10**.

(12) FIG. 2 shows the connector module **100** in a disassembled state where the connector **20** is not inserted into the adapter **10**. In the view illustrated by FIG. 2 a top surface **14** of the adapter **10** is seen to be flat other than some exemplary notches at two of the corners. The adapter **10** also includes two opposing side walls **19**, as well as a front side surface **31**. The front side surface **31** includes an opening **12** through which a prong housing **26** of the connector **20** may be inserted through when the connector is inserted into the adapter **10** via a rear side. The front side surface **31** also includes a slot **17** at a bottom end of the opening **12** that provides a space that may be utilized to better insert the connector **20** into the adapter **10**, and/or remove the connector **20** from the adapter **10**. The front side surface **31** also includes an icon pocket **32** where an icon piece may fit into, and or an adhesive backed label may be placed. A head-on front side view of the adapter **10** is also illustrated in FIG. 5.

(13) From the perspective view illustrated in FIG. 2, the connector **20** is shown to include the prong housing **26** which is a protrusion that is inserted through the adapter **10** to stick out, at least in part, past the opening **12** of the adapter **10**. The prong housing **26** includes an inner chamber housing two prongs **23** for mating to corresponding mating components of a plug.

(14) The connector **20** also includes two detents **24**, one on each opposite side wall **27** of the prong housing **26**. When the connector **20** is inserted into an inner cavity **33** of the adapter **10**, protrusions **18** within the inner cavity **33** are configured to slide and fit into respective detents **24** on the side walls **27** of the prong housing **26**. A rear side perspective view of the adapter **10** is illustrated in FIG. 4 and a rear side head-on view of the adapter **10** is illustrated in FIG. 6, where the protrusions **18** are shown to be positioned on opposite sides of the inner cavity **33**. The fitting of the protrusions **18** from the inner cavity **33** of the adapter **10** into the detents **24** of the connector **20** aids in providing a more secure installation and attachment of the connector **20** to the adapter **10** when assembling the connector module **100**.

(15) FIG. 3 shows a bottom side perspective view of the adapter **10** and the connector **20**. A bottom surface **15** of the adapter **10** includes a thinner surface area **15-2** where a material (e.g., plastic) of the bottom surface **15** has been thinned out, and a front latching feature **15-3** for snapping the adapter **10** into a panel or faceplate. The bottom surface **15** also includes an extension portion **15-1** that extends beyond a plane of the top surface **14**. The extension portion **15-1** includes a latch **13** for securing the connector **20** within the adapter **10** by engaging an edge **25** of the connector **20** when the connector **20** is inserted into the adapter **10**. The bottom surface **15** also includes two holes **11** that are configured to line up with two holes **21** on the connector **20** when the connector is inserted into the adapter **10** into the assembled state. A screw is configured to be inserted and rotated into the holes **11**, **21** to screw down and secure wire conductors that are being housed within the connector **20** after being inserted into the connector **20** via the wire openings **22**.

(16) From the rear side views into the inner cavity **33** of the adapter **10** illustrated in FIGS. 4 and 6, two fit extensions **16** are shown to extend up from the bottom surface **15**. The fit extensions **16** are positioned and sized to provide a firm fit of the connector **20** into the inner cavity **33** of the adapter **10** when the connector module is in the assembled state. According to other embodiments, one or more fit extensions **16** may be included within the inner cavity **33**. For example, a single fit

extension having a greater width may be used in place of the two fit extensions **16** shown in FIG. **4** according to other embodiments. As another example, one or more additional fit extensions **16** may be further added to the two fit extensions **16** shown in FIG. **4** according to other embodiments.

(17) FIG. **7** shows the connector module **100** in the assembled state and prepared to receive a corresponding plug **200**. The plug **200** includes a mating extension **203** configured to be inserted into the prong housing **26** of the connector **20**, where mating components inside the mating extension **203** of the plug **200** are configured to mate with the two prongs **23** inside the prong housing **26** of the connector **20** to result in an electronic coupling between the plug **200** and the connector **20**. FIG. **8** shows the plug **200** being mated to the connector **20** in the assembled connector module **100**.

(18) The plug **200** includes two wire openings **201** at a rear surface, where each of the wire openings **201** are configured to receive a respective wire conductor from a single pair of wires (e.g., single pair cable). As provided by the exemplary installation of the connector module **100** into network equipment **300** shown in FIG. **9**, a first wire **41** and a second wire **42** from a single pair cable may be terminated and secured into the plug **200** by twisting down screws into two holes **202** in the plug **200**. Then the plug **200** may be simply mated to the corresponding connector **20** that has already been attached to the network equipment **300** using the adapter **10** portion of the connector module **100**. So the adapter **10** is configured to have the predetermined size, shape, and coupling mechanisms to fit into a port opening **301** in the network equipment **300** and be secured within the port opening **301**.

(19) In FIG. **9** the network equipment **300** is shown to be a patch panel, but other types of network equipment that may be used for communicating using single pair cables may also be applicable for utilizing the connector module **100**. In particular, FIG. **9** shows the connector module **100** installed into a port opening **301** provided on the network equipment **300**, where the connector module **100** includes a Panduit Mini-Com adapter and the network equipment **300** is a patch panel where the port opening **301** has characteristics to fit the Panduit Mini-Com adapter.

(20) It follows that the connector module **100** offers a simple and effective solution for terminating and coupling single pair cables in various use-case environments. For example, the single pair cables may be used for transmitting power using pulsed power systems to power components in a DAS installation environment.

(21) Furthermore, while the particular embodiments described herein have been shown and described, it will be obvious to those skilled in the art that changes and modifications may be made without departing from the teaching described herein with reference to the connector module **100**. The matter set forth in the foregoing description and accompanying drawings is offered by way of illustration only and not as limitation. The actual scope of the disclosure is intended to be defined in the following claims when viewed in their proper perspective.

## Claims

1. A connector module for installing into an equipment opening, the connector module comprising: an adapter configured to receive a connector jack, the adapter comprising: a top surface; a first side wall; a second side wall opposite the first side wall; a front side surface including an opening for receiving the connector jack; and a bottom surface including a pair of adapter screw holes configured to receive a pair of screws; and wherein an internal cavity for housing the connector jack is formed by at least a portion of the top surface, a portion of the first side wall, a portion of the second side wall, a portion of the front side surface, and a portion of the bottom surface, wherein the connector jack includes a pair of connector screw holes configured to line up with the adapter pair of screw holes when the connector jack is housed within the internal cavity.
2. The connector module of claim 1, the front side surface further including a slot located at a bottom end of the opening.

3. The connector module of claim 1, the internal cavity comprising: at least one fit extension configured to adapt the connector jack inside the internal cavity.
4. The connector module of claim 1, the internal cavity comprising: at least one protrusion, wherein the at least one protrusion is configured to fit within a respective detent included on the connector jack.
5. The connector module of claim 1, wherein the adapter is configured to snap fit into a pre-determined opening included in a networking equipment.
6. The connector module of claim 5, wherein the networking equipment is a patch panel.
7. The connector module of claim 1, wherein the connector jack includes two conductor prongs for engaging corresponding mating components of a connector plug inserted into the connector jack.
8. The connector module of claim 1, wherein the connector jack includes two wire openings configured to lead into respective wire housings, each wire opening configured to receive a respective wire conductor from a single pair of conductors such that at least a portion of each wire conductor is housed within the respective wire housings, and wherein the pair of screws are configured to fasten down through the adapter pair of screw holes and the connector pair of screw holes to clamp down on the wire conductors housed within the respective wire housings.
9. The connector module of claim 8, wherein the respective wire conductor is 12 AWG or higher.
10. The connector module of claim 1, wherein when the connector jack is housed inside the internal cavity, the opening of the front side faces perpendicular to an opening for screw holes included on a bottom surface of the connector jack.
11. A connector jack comprising: a prong housing forming an inner chamber, the inner chamber including at least two prongs; a first side wall, the first side wall including a first detent configured to receive a first protrusion included in an inner cavity of an adapter when the connector jack is installed into the adapter; a second side wall, the second side wall including a second detent configured to receive a second protrusion included in the inner cavity of the adapter when the connector jack is installed into the adapter; a pair of connector screw holes located on a bottom surface of the connector jack, wherein the pair of connector screw holes are configured to line up to a corresponding pair of adapter screw holes on a bottom surface of the adapter when the connector jack is installed into the adapter such that a pair of screws are configured to pass through both the pair of connector screw holes and the pair of adapter screw holes when the connector jack is installed into the adapter; and at least two wire openings configured to receive a respective wire conductor from a single pair of conductors.
12. The connector jack of claim 11, wherein an opening for the screw holes are perpendicular to an opening of the inner chamber.
13. The connector jack of claim 11, wherein the at least two wire openings face a different direction from an opening of the inner chamber.
14. The connector jack of claim 11, wherein the at least two wire openings are configured to receive 12 AWG or higher wire conductors.
15. The connector jack of claim 11, wherein the conductor jack is configured to be installed into an adapter, the adapter comprising: a top surface; a first side wall; a second side wall opposite the first side wall; a front side surface including an opening for receiving the connector jack; and wherein an internal cavity for housing the connector jack is formed by at least a portion of the top surface, a portion of the first side wall, a portion of the second side wall, a portion of the front side surface, and a portion of the bottom surface.
16. The connector jack of claim 11, wherein each wire opening is configured to lead into respective wire housings, and each wire opening is configured to receive the respective wire conductor from the single pair of conductors such that at least a portion of each wire conductor is housed within the respective wire housings, and wherein the pair of screws are configured to fasten down through the adapter pair of screw holes and the connector pair of screw holes to clamp down on the wire conductors housed within the respective wire housings.

